Teacher Guide — senior / module-11-multiplication-near-base

IKS Curriculum

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Overview

Multiplication Near a Power of 10 — Nikhilam

Band: Senior (Grades 9–12) · **Sessions:** 10 × 45 min · **Sanskrit:** Nikhilaṁ Navataścaramaṁ Daśataḥ (Multiplication)

Overview

When both factors are within a few units of a power of 10 (e.g. 97 × 96, 103 × 104), Tirthaji's Nikhilam sūtra gives a two-step mental method that is much faster than long multiplication. Uses deficits/excesses from a base, with cross-subtraction on one side and product-of-deficits on the other.

NEP 2020 alignment

- §4.6 — Integration of Indian Knowledge Systems
- §4.23 — Experiential learning
- §4.27 — Local context and language

Key terms (this module)

- *Nikhilaṁ* — the Nikhilam sūtra (also used in Module 9 for subtraction)
- *vinkulam* — deficit (number below base)
- *ādhikya* — excess (number above base)
- *Ūrdhva-tiryagbhyāṁ* — "vertically and crosswise" — the general-purpose sūtra

Prerequisites

Module 2 (Panchabhūta) is recommended before this module. Modules 5 (Bindu Paddhati / dot addition) and 7 (×11 Multiplication) are useful prerequisites for Module 9 and 11.

Materials needed (across 10 days)

- Notebook + pencils per student
- Whiteboard / chart paper

- (See `lesson-plan.md` for per-day material lists)

10-day map

- **Day 1** — Finger-Trick Prelude: Practice the below-10 finger method (e.g., 7 × 8 using deficits).
- **Day 2** — Two-Digit × Two-Digit Near 100 (Below): Compute 97 × 96 using deficits.
- **Day 3** — Two-Digit × Two-Digit Near 100 (Above): Compute 103 × 104 using excesses.
- **Day 4** — Mixed — One Above, One Below: Compute 102 × 98 with sign care.
- **Day 5** — 3-Digit × 3-Digit Near 1000: Compute 997 × 996 mentally.
- **Day 6** — Why It Works — The Algebra: Derive $(B-a)(B-b) = B(B-a-b) + ab$.
- **Day 7** — When To Use It: Identify problems where Nikhilam is faster than long multiplication.
- **Day 8** — Speed Sprints: Race traditional vs Nikhilam on 30 problems.
- **Day 9** — Mixed Practice + Module 7 Crossover: Solve problems combining ×11 and near-base methods.
- **Day 10** — Final Assessment: Summative quiz and method-comparison reflection.

Files in this pack

- `lesson-plan.md` — detailed 10-day arc
- `sources.md` — sources cited in this module
- `teacher-notes.md` — common misconceptions, differentiation
- `parent-guide.md` — what parents should know
- `days/day-NN-*.md` — 10 per-day lesson plans
- `quizzes/formative.md`, `quizzes/summative.md`
- `activities/activity-01-*.md`
- `assessments/rubric.md`, `assessments/project-brief.md`
- `student-workbook.md`
- `writing-prompts.md` (or `oral-recall.md` for Primary)
- `homework.md`
- `slides/deck.md`

Voice

Analytical, comparative, primary-source aware; 20–30 words on average. Pitches the concept first, frames the tradition as historical context, and uses IAST for all Sanskrit terms.

10-Day Lesson Plan

Lesson Plan — Multiplication Near a Power of 10 — Nikhilam (Senior)

10-day arc, 45 min per session.

Day 1 — Finger-Trick Prelude

Objective: Practice the below-10 finger method (e.g., 7 × 8 using deficits).

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 2 — Two-Digit × Two-Digit Near 100 (Below)

Objective: Compute 97 × 96 using deficits.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 3 — Two-Digit × Two-Digit Near 100 (Above)

Objective: Compute 103 × 104 using excesses.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 4 — Mixed — One Above, One Below

Objective: Compute 102 × 98 with sign care.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 5 — 3-Digit × 3-Digit Near 1000

Objective: Compute 997 × 996 mentally.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 6 — Why It Works — The Algebra

Objective: Derive $(B-a)(B-b) = B(B-a-b) + ab$.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 7 — When To Use It

Objective: Identify problems where Nikhilam is faster than long multiplication.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 8 — Speed Sprints

Objective: Race traditional vs Nikhilam on 30 problems.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 9 — Mixed Practice + Module 7 Crossover

Objective: Solve problems combining ×11 and near-base methods.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 10 — Final Assessment

Objective: Summative quiz and method-comparison reflection.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Pacing notes

- Senior sessions are 45 min. Adjust if your block is different.
- Days 5 and 9 are buffer days — use to reinforce weaker concepts or extend stronger ones.
- The summative project (see `assessments/project-brief.md`) runs in parallel from Day 6 onward.

Differentiation

- **For advanced students:** see `teacher-notes.md` for extension prompts.
- **For students needing more support:** simplify activities, do more pair-work, allow oral instead of written responses.

Teacher Notes

Teacher Notes — Multiplication Near a Power of 10 — Nikhilam (Senior)

Common student misconceptions

- *"This is religious, not scientific."* — Frame the tradition as historical observation. Pañcabhūta-like classifications of matter exist in many cultures. Indian astronomy and math are documented, peer-reviewed historical fields.
- *"The Sanskrit terms are just jargon."* — Insist on the triple format (Sanskrit / IAST / meaning) once, then use the Sanskrit alone. The terms are precise; the discipline of using them properly is itself a learning outcome.
- *"If it's traditional, it must work."* — Distinguish empirical claims (testable) from ethical/aesthetic frames (which need separate justification). Encourage students to ask "how would I test this?".

Differentiation strategies

- **Extension for advanced students:** offer one optional "deeper read" per day — usually a primary source or modern paper.
- **Support for students needing scaffolds:** more pair-work, allow oral responses, provide partial outlines for writing prompts.
- **Multilingual classrooms:** Sanskrit terms are universal; vernacular equivalents (Hindi, Tamil, Telugu, Bengali, Marathi etc.) should be welcomed.

Safety

- Any activity involving food, plants, fire, or outdoor work needs explicit safety briefing.
- See the relevant day file for activity-specific safety notes.

Honest framing

- Where modern science clearly extends or revises traditional claims, say so.
- Where the tradition's framing is best understood as ethical/aesthetic, present it as such (not as competing science).
- Where attribution is debated (e.g., the "Vedic" status of Tirthaji's mathematics), present the debate honestly.

End-of-module reflection (for teachers)

- Which concepts landed well? Which didn't?
- Which students grew most? Which need a different approach next module?
- Add notes to `teacher-notes.md` for the next teacher who uses this pack.

Sources

Sources — Multiplication Near a Power of 10 — Nikhilam (Senior)

Primary and secondary sources

- Bharati Krishna Tirthaji, *Vedic Mathematics* (1965), chapter on multiplication
- S. G. Dani, "Myths and reality: On Vedic mathematics," *Frontline* (1993)
- NCERT Mathematics Class VII, ch. 1 (Integers, including multiplication)

A note on citation discipline

Per the curriculum's style guide, citations must be verifiable. If a verse number, page, or attribution cannot be confirmed, the source is paraphrased without invented attribution. Where direct quotes appear in Senior-band day files, the source is identified in IAST + edition.

For deeper reading

- The shared glossary at `curriculum/_shared/glossary.md` contains the IAST + meaning for every Sanskrit term used.
- Other modules in this curriculum that connect: see the module README's "Prerequisites" section.

Day 1 — Finger-Trick Prelude

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: practice the below-10 finger method (e.g., 7 × 8 using deficits).

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilaṁ* (IAST: Nikhilaṁ; lit. "the Nikhilam sūtra (also used in Module 9 for subtraction)")

Primary source (today's reading)

From *Bharati Krishna Tirthaji* — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *"What's one thing you remember about yesterday's lesson?"*
- Hook: introduce a phenomenon or question that motivates today's concept.

Core (20 min)

1. **Recall Module 9.** Nikhilam = "all from 9, last from 10" — the complement to a base. Same sūtra, NEW application: multiplication.

2. **The below-10 finger demo.** Each hand has 5 fingers; together 10. To multiply 7 × 8:

   - Curl 7−5=2 fingers on left hand.
   - Curl 8−5=3 fingers on right hand.
   - Curled fingers in total: 5 → that's the tens place: 50.
   - Uncurled fingers (left × right): 3 × 2 = 6 → that's the units: 6.

    - Answer: 56. ✓
3. **Why this works.** Algebraically: $(10 - a)(10 - b) = 100 - 10(a+b) + ab = 10(10-(a+b)) + ab$. The "curled fingers" count is $(a+b)-5$ — wait, let me re-derive. Actually: curled = $(10-a - 5) + (10-b - 5)$ wait… let me restate cleanly. For 7×8: $7 = 5+2$ (uncurled fingers on left), $8 = 5+3$ (right). The formula: tens digit = uncurled-left + uncurled-right − 5; units = uncurled-left × uncurled-right. Let's just use the Nikhilam algebra in next lesson.

4. **What we'll learn this module.** A general method: when factors are CLOSE to a base (10, 100, 1000), Nikhilam-multiplication is much faster than long multiplication.

5. **Speed test.** Compute 8×9 mentally. Compare your method with traditional times table. Most students will use memorisation — the next lessons show why Nikhilam is faster for LARGER near-base products.

Activity (15 min)

A scaled version of *Method Speed Race* (see `activities/`). Today's slice: focus on the part of the activity that reinforces Finger-Trick Prelude.

Wrap-up (5 min)

- One-sentence exit ticket: *"What's one thing you learned today?"*
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 2 — Two-Digit × Two-Digit Near 100 (Below)

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute 97×96 using deficits.

Materials

- Whiteboard / chart paper

- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *vinkulam* (IAST: vinkulam; lit. “deficit (number below base)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *“What’s one thing you remember about yesterday’s lesson?”*
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The two-step rule for both factors below 100.**
    - Find deficits from 100: e.g., 97 → −3, 96 → −4.
    - **Left part:** cross-subtract. 97 − 4 = 93 (or equivalently 96 − 3 = 93).
    - **Right part:** product of deficits. 3 × 4 = 12.
    - Combine: **9312**.
2. **Worked examples.**
    - 97 × 96 = 93 | 12 = **9312**.
    - 88 × 95 = 83 | 60 = **8360**. (Deficits: 12, 5. Cross: 88−5=83 or 95−12=83. Product: 12×5=60.)
    - 99 × 99 = 98 | 01 = **9801**. (Deficits 1, 1. Cross: 98. Product: 1. Pad to 2 digits: 01.)
3. **Why the padding matters.** The right part must have as many digits as the base has zeros (here, 2 zeros → 2-digit right part). If product is single-digit, pad with leading zero. If product overflows (3+ digits), carry to left part.
4. **Algebra.** Base B = 100. Factor 1 = B − a, factor 2 = B − b. Then (B−a)(B−b) = B(B−a−b) + ab = B·(cross) + (right). The cross-subtract IS B − a − b. The product of deficits IS ab.
5. **Independent practice.** 8 problems, all below-100 case. 10 minutes.

Activity (15 min)

A scaled version of *Method Speed Race* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces Two-Digit × Two-Digit Near 100 (Below).

Wrap-up (5 min)

- One-sentence exit ticket: *“What’s one thing you learned today?”*
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 3 — Two-Digit × Two-Digit Near 100 (Above)

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute 103 × 104 using excesses.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *ādhikya* (IAST: ādhikya; lit. “excess (number above base)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *“What’s one thing you remember about yesterday’s lesson?”*
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Above-100 variant.** Use *excesses* instead of deficits.
    - 103 = 100 + 3 → excess +3.
    - 104 = 100 + 4 → excess +4.
2. **The two-step rule for above-base.**
    - **Left part:** cross-add. 103 + 4 = 107 (or 104 + 3 = 107).

    - **Right part:** product of excesses. 3 × 4 = 12.
    - Combine: **10712**.
3. **Worked examples.**
    - 103 × 104 = 107 | 12 = **10712**.
    - 110 × 105 = 115 | 50 = **11550**.
    - 102 × 109 = 111 | 18 = **11118**.
4. **Algebra (above-base).** (B+a)(B+b) = B(B+a+b) + ab. Cross-ADD = B+a+b. Product of excesses = ab.
5. **Symmetry recognition.** Below-base: cross-SUBTRACT. Above-base: cross-ADD. Both: product-of-difference-from-base. *“This symmetry hints at the general pattern — tomorrow we’ll see the mixed case.”*

Activity (15 min)

A scaled version of *Method Speed Race* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces Two-Digit × Two-Digit Near 100 (Above).

Wrap-up (5 min)

- One-sentence exit ticket: *“What’s one thing you learned today?”*
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 4 — Mixed — One Above, One Below

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute 102 × 98 with sign care.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Ūrdhva-tiryagbhyāṁ* (IAST: Ūrdhva-tiryagbhyāṁ; lit. ““vertically and crosswise” — the general-purpose sūtra”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *“What’s one thing you remember about yesterday’s lesson?”*
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Mixed case (one above, one below).** Care with signs! Example: 102 × 98.
    - 102 → +2 (excess), 98 → −2 (deficit).
    - Left part: 102 − 2 = 100 (or 98 + 2 = 100).
    - Right part: (+2)(−2) = **−4**.
    - Combine: 100 | −4. But how to interpret −4 in the right slot?
2. **The “borrow back” trick.** Negative right part → subtract from left. 100|−4 means 10000 − 4 = **9996**.
3. **Alternative algebra check.** $102 \times 98 = (100+2)(100-2) = 100^2 - 4 = 9996$. ✓ (Difference of squares identity.)
4. **General mixed-case rule.** Cross-subtract (using the deficit sign) for left; multiply excess and deficit (negative result) for right; convert negative right to a borrow from left.
5. **Independent practice.** 5 problems mixing all three cases (both below, both above, mixed). 10 minutes.

Activity (15 min)

A scaled version of *Method Speed Race* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces Mixed — One Above, One Below.

Wrap-up (5 min)

- One-sentence exit ticket: *“What’s one thing you learned today?”*
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 5 — 3-Digit × 3-Digit Near 1000

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute 997 × 996 mentally.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilaṁ* (IAST: Nikhilaṁ; lit. "the Nikhilam sūtra (also used in Module 9 for subtraction)")

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *"What's one thing you remember about yesterday's lesson?"*
- Hook: introduce a phenomenon or question that motivates today's concept.

Core (20 min)

1. **Extension to base 1000.** Same rule, base now 1000.

    - 997 → deficit 3.
    - 996 → deficit 4.
    - Left: 997 − 4 = 993.
    - Right: 3 × 4 = 12. Must be 3 digits (because base has 3 zeros). Pad: **012**.

- Combine: **993012**.

2. **Above 1000.** 1003 × 1004 = 1007 | 012 = **1007012**.
3. **Mixed case.** 1003 × 997 = (1000+3)(1000−3) = 10^6 − 9 = 999991. (Difference of squares.)
4. **Pattern recognition.** The base extends naturally: 10, 100, 1000, 10000… Each next-decade-base just shifts everything. The CORE rule (cross-subtract + product-of-deficits) is identical.
5. **Independent practice.** 5 near-1000 problems. The challenge: keeping the right-part digit count correct.

Activity (15 min)

A scaled version of *Method Speed Race* (see `activities/`). Today's slice: focus on the part of the activity that reinforces 3-Digit × 3-Digit Near 1000.

Wrap-up (5 min)

- One-sentence exit ticket: *"What's one thing you learned today?"*
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 6 — Why It Works — The Algebra

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: derive (b−a)(b−b) = b(b−a−b) + ab.

Materials

- Whiteboard / chart paper
- Notebook, pen per student

- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *vinkulam* (IAST: vinkulam; lit. "deficit (number below base)")

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *"What's one thing you remember about yesterday's lesson?"*
- Hook: introduce a phenomenon or question that motivates today's concept.

Core (20 min)

1. **Derive the formula.** Let B = the base (10, 100, 1000). Factor 1 = B − a, Factor 2 = B − b.

   $(B - a)(B - b) = B^2 - Ba - Bb + ab = B(B - a - b) + ab = B \cdot (\text{left part}) + (\text{right part})$

   So the left part is $B - a - b = (B - a) - b = (B - b) - a$ = either factor minus the OTHER's deficit. The right part is just ab.

2. **Why "cross-subtract"?** B − a − b = first factor − (B − second factor) = first factor − second's deficit. The cross-subtraction is just rearranging the algebra into a mental-math procedure.

3. **Above-base proof.** $(B + a)(B + b) = B^2 + Ba + Bb + ab = B(B + a + b) + ab$. Cross-add, product of excesses.

4. **Mixed case proof.** $(B + a)(B - b) = B^2 - Bb + Ba - ab = B(B + a - b) - ab$. The "−ab" in the right slot is what causes the sign issue.

5. **Connecting the dots.** This is just basic high-school algebra — no mystery. Tirthaji's sūtra system gives the procedure a memorable name; the algebra gives it the proof.

Activity (15 min)

A scaled version of *Method Speed Race* (see `activities/`). Today's slice: focus on the part of the activity that reinforces Why It Works — The Algebra.

Wrap-up (5 min)

- One-sentence exit ticket: *"What's one thing you learned today?"*
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.

- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 7 — When To Use It

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify problems where nikhilam is faster than long multiplication.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *ādhikya* (IAST: ādhikya; lit. "excess (number above base)")

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *"What's one thing you remember about yesterday's lesson?"*
- Hook: introduce a phenomenon or question that motivates today's concept.

Core (20 min)

1. **The decision tree.**
    - Both factors within ~5 of the same base → Nikhilam is fast.
    - One factor close to base, other far → traditional multiplication often faster.
    - Both factors far from any base (e.g. 47 × 63) → traditional.
    - Both factors near DIFFERENT bases (e.g. 98 × 1005) → tricky; pick a common base and scale.
2. **Speed comparison.** Time 97 × 96 vs 47 × 63 mentally. Most students will find Nikhilam dominates the first; long multiplication dominates the second.
3. **Pedagogical limit.** Nikhilam multiplication is NOT a universal speed-up. It's a tool for ONE problem class. Recognising when to use it is the skill.

4. **Practice with strategic choice.** 10 problems, mixed. Each student picks the best method and notes it next to the answer.

5. **Reflection.** *"What pattern of problems makes Nikhilam shine? What pattern defeats it?"* Each student writes 3 sentences.

Activity (15 min)

A scaled version of *Method Speed Race* (see `activities/`). Today's slice: focus on the part of the activity that reinforces When To Use It.

Wrap-up (5 min)

- One-sentence exit ticket: *"What's one thing you learned today?"*
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 8 — Speed Sprints

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: race traditional vs nikhilam on 30 problems.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Ūrdhva-tiryagbhyāṁ* (IAST: Ūrdhva-tiryagbhyāṁ; lit. ""vertically and crosswise" — the general-purpose sūtra")

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *"What's one thing you remember about yesterday's lesson?"*
- Hook: introduce a phenomenon or question that motivates today's concept.

Core (20 min)

1. **Sprint A (Nikhilam-favourable, 10 problems, 4 min).** All factors within 5 of 100 or 1000.

2. **Sprint B (mixed, 10 problems, 4 min).** Mix of near-base and far-from-base.

3. **Sprint C (long-mult-favourable, 10 problems, 4 min).** All factors far from base.

4. **Analysis.** Each student records times. Most: A < B < C dramatically. The CONCRETE evidence that Nikhilam works for ONE class.

5. **Discussion.** *"What's your strategy choice now? When will you use Nikhilam in real life?"* Each student writes 2 sentences.

Activity (15 min)

A scaled version of *Method Speed Race* (see `activities/`). Today's slice: focus on the part of the activity that reinforces Speed Sprints.

Wrap-up (5 min)

- One-sentence exit ticket: *"What's one thing you learned today?"*
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 9 — Mixed Practice + Module 7 Crossover

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: solve problems combining ×11 and near-base methods.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilaṁ* (IAST: Nikhilaṁ; lit. “the Nikhilam sūtra (also used in Module 9 for subtraction)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *“What’s one thing you remember about yesterday’s lesson?”*
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Crossover with Module 7.** Each student solves 5 problems combining ×11 method + Nikhilam:

    - 11 × 99 = 1089 (use ×11)
    - 97 × 96 = 9312 (use Nikhilam)
    - 11 × 98 = 1078 (use ×11)
    - 105 × 95 = 9975 (use difference-of-squares or Nikhilam mixed)
    - 11 × 87 = 957 (use ×11 with carry)
2. **Strategy reflection.** Pair-discuss: *“For each, what method DID you use? Was it the fastest?”*

3. **Build a personal toolkit.** Each student updates their “personal mental-math toolkit” page (started in Module 7) to include Nikhilam multiplication.

4. **One more example.** Solve 998 × 997 mentally. (Nikhilam: deficits 2, 3 → left = 995, right = 006 → **995006**.) 4 seconds with the trick; >1 minute by long multiplication.

5. **Reflection.** *“What’s the most useful thing you learned this module?”*

Activity (15 min)

A scaled version of *Method Speed Race* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces Mixed Practice + Module 7 Crossover.

Wrap-up (5 min)

- One-sentence exit ticket: *“What’s one thing you learned today?”*
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 10 — Final Assessment

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: summative quiz and method-comparison reflection.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *vinkulam* (IAST: vinkulam; lit. “deficit (number below base)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *“What’s one thing you remember about yesterday’s lesson?”*
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Method-comparison written reflection (10 min).** *“In 300 words, compare Tirthaji’s near-base multiplication with traditional long multiplication. When does each excel? What does Tirthaji’s method teach about number sense?”*

2. **Summative quiz (25 min).** Mix of Nikhilam (below, above, mixed), algebraic proof, strategy choice. See `quizzes/summative.md`.

3. **Personal mental-math toolkit share (10 min).** Each student shows their one-page toolkit.

4. **Module close.** *"One number-sense habit you'll carry forward?"*

5. **Linkage.** Module 14 (Magic Squares) will use related place-value insights for grid arithmetic.

Activity (15 min)

A scaled version of *Method Speed Race* (see `activities/`). Today's slice: focus on the part of the activity that reinforces Final Assessment.

Wrap-up (5 min)

- One-sentence exit ticket: *"What's one thing you learned today?"*
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Formative Quiz — Multiplication Near a Power of 10 — Nikhilam (Senior)

When: mid-module (after Day 5). **Duration:** 25 minutes. **Format:** mix of short answer, multiple choice, and reflection.

Questions

1. Name the three / four key terms introduced in this module so far. Give the IAST and one-line meaning of each.
2. State the central idea of Multiplication Near a Power of 10 — Nikhilam in one sentence.
3. (Multiple choice) Which day in this module focused on 'Two-Digit × Two-Digit Near 100 (Above)'? (a) Day 1 (b) Day 2 (c) Day 3 (d) Day 4

4. True / False — and explain in one sentence: *Multiplication Near a Power of 10 — Nikhilam is purely a religious concept and has no scientific framing.*
5. Connect Multiplication Near a Power of 10 — Nikhilam to **one** other module in this curriculum. Which module, and how?
6. List **two** sources cited so far in this module.
7. (Short answer) Why does the module distinguish 'Nikhilaṁ' from related concepts? Give one example.
8. (Short answer) In your own words, explain Day 3's concept.
9. Identify **one** common misconception about Multiplication Near a Power of 10 — Nikhilam and rebut it in one sentence.
10. Write **one** question you have about this module so far — something you want clarified before the summative.

How this is used

This quiz is **formative** — it's a check-in to see what's landing, not a grade. Teacher reviews answers to adjust pacing for Days 6–10. Students may swap and self-mark.

Summative Quiz — Multiplication Near a Power of 10 — Nikhilam (Senior)

When: end of module (after Day 10). **Duration:** 45 minutes. **Total points:** 30.

Questions

Part A — Vocabulary (~5 points) A1. Define *Nikhilaṁ* in your own words. A2. Define *vinkulam* in your own words. A3. Define *ādhikya* in your own words. A4. Define *Ūrdhva-tiryagbhyāṁ* in your own words.

Part B — Concepts (~10 points) B1. Explain the central idea of Multiplication Near a Power of 10 — Nikhilam in your own words (3–4 sentences). B2. Describe the method or framework students used in Day 4 of this module. B3. Identify ONE source cited in this module and explain why it matters. B4. Compare Multiplication Near a Power of 10 — Nikhilam with a concept from another module (your choice). What's similar, what's different? B5. Identify ONE common misconception about this module and rebut it in 2–3 sentences.

Part C — Application (~5 points) C1. Give an example of Multiplication Near a Power of 10 — Nikhilam from your own daily life. C2. Describe one way this module connects to a current real-world issue (or scientific question).

Part D — Reflection (~5 points; ungraded for content) D1. What surprised you most in this module? D2. What's one question you still have?

Part E — Primary source (~5 points; Senior only) E1. Quote or paraphrase one passage from a primary source cited in this module. Explain it in your own words.

Marking guidance

- Vocabulary: give credit for correct meaning, even if exact wording differs.
- Concepts: look for evidence of understanding, not memorisation.
- Application: any reasonable example accepted.
- Reflection: ungraded for content; awarded for thoughtfulness.

Pass mark

A score of 18/30 indicates the student has met the module’s learning objectives.

Homework

Homework — Multiplication Near a Power of 10 — Nikhilam (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — Finger-Trick Prelude** · Write 2 sentences about today’s concept. (~10 min)
2. **Day 2 — Two-Digit × Two-Digit Near 100 (Below)** · Write 2 sentences about today’s concept. (~10 min)
3. **Day 3 — Two-Digit × Two-Digit Near 100 (Above)** · Write 2 sentences about today’s concept. (~10 min)
4. **Day 4 — Mixed — One Above, One Below** · Write 2 sentences about today’s concept. (~10 min)
5. **Day 5 — 3-Digit × 3-Digit Near 1000** · Write 2 sentences about today’s concept. (~10 min)
6. **Day 6 — Why It Works — The Algebra** · Write 2 sentences about today’s concept. (~10 min)
7. **Day 7 — When To Use It** · Write 2 sentences about today’s concept. (~10 min)
8. **Day 8 — Speed Sprints** · Write 2 sentences about today’s concept. (~10 min)
9. **Day 9 — Mixed Practice + Module 7 Crossover** · Write 2 sentences about today’s concept. (~10 min)
10. **Day 10 — Final Assessment** · Write 2 sentences about today’s concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.

Writing Prompts

Writing Prompts — Multiplication Near a Power of 10 — Nikhilam (Senior)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 300-word response.

1. In 3–4 sentences, explain Multiplication Near a Power of 10 — Nikhilam to a friend who hasn't taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Multiplication Near a Power of 10 — Nikhilam with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Multiplication Near a Power of 10 — Nikhilam? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?
9. Read one primary-source passage cited in this module. Translate or paraphrase it; comment on what it adds to your understanding.
10. Identify a contemporary debate about the IKS framework presented here (e.g., is Tirthaji's mathematics genuinely Vedic? are doṣas a valid framework today?). Take a position with two pieces of evidence.

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Activity 1 — Method Speed Race

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Duration:** 20 min

Purpose

Reinforce the day-by-day concepts of Multiplication Near a Power of 10 — Nikhilam through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

20 problem cards, stopwatches, two columns on whiteboard

What students do

Split class in two: one team solves with traditional long multiplication, the other with Nikhilam. Race on identical problems. Discuss which problems favour which method.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (10 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Project Brief

Project Brief — Personal Mental-Math Toolkit

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior

What you will do

Each student compiles a one-page toolkit: when to use ×11, when Nikhilam multiplication, when traditional long-mult. Include 3 worked examples for each method.

Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.

- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a poster / model / digital deliverable).
- **Written reflection** (500 words (Senior)): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (5-minute share with the class).

Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

Sources

You must cite at least 3 sources. Use the form: *Author*, *Title*, *year*. (See `../sources.md` for examples.)

Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.

Rubric

Assessment Rubric — Multiplication Near a Power of 10 — Nikhilam (Senior)

This rubric applies to the project (see `project-brief.md`) and may also inform the summative quiz.

Criteria

Accuracy of concepts — uses Sanskrit terms correctly, gets the central idea right

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.

- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Source discipline — cites sources, distinguishes traditional framing from empirical claim

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Application — connects the module to other modules and to real-world examples

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Presentation — clear, organised, evidence-supported

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Reflection — honest, thoughtful, identifies what was learned and what remains uncertain

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

How the teacher uses this rubric

1. After the project share, score each student on each criterion.
2. Convert to a band (4 = Exceeding, 3 = Meeting, etc.) for the report card.
3. Share at least one strength and one area for growth with each student.

Self-assessment (Senior students)

Senior students complete a parallel self-assessment using the same rubric before the teacher's assessment. The teacher and student then meet to compare.